

Computational Work 2: Cholesky Factorization

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Course: Computational Matrix Methods

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Abstract

This work investigates two computational problems involving numerical linear algebra and the numerical solution of a one-dimensional steady-state heat conduction problem. First, a symmetric matrix generated according to the prescribed procedure is analyzed to determine whether it is positive definite, followed by the computation of its Cholesky factorization, condition number, and the solution of the associated linear system. Second, the steady-state heat conduction equation in a unit-length bar is discretized using the finite difference formulation specified in the problem statement, resulting in a linear system whose solution provides the temperature distribution along the bar. The numerical results are analyzed with respect to the algebraic properties of the matrices, the conditioning of the linear systems, and the physical interpretation of the computed temperature profile. All problem-specific data are generated deterministically from the student's enrollment number according to the procedures provided in the assignment.

1 Problem

The computational work is divided into two problems involving numerical linear algebra and the numerical solution of a steady-state heat conduction problem.

1.1 Problem 1: Symmetric Positive Definite Matrix

Let $A \in \mathbb{R}^{10 \times 10}$ and $b \in \mathbb{R}^{10}$ be generated according to the procedure given in Code 1 of Appendix A.1, using the student's enrollment number as the random seed.

The following tasks are considered:

1. Determine whether the matrix A is symmetric positive definite. If so, compute its Cholesky factorization

$$A = GG^T,$$

and present the Cholesky factor G .

2. Compute the condition number $\kappa(A)$ and justify the result.
3. Solve the linear system

$$Ax = b.$$

1.2 Problem 2: One-Dimensional Steady-State Heat Conduction

Consider the one-dimensional steady-state heat conduction problem in a unit-length bar,

$$-k \frac{d^2 T}{dx^2} = f, \quad 0 \leq x \leq 1, \quad (1)$$

where $T(x)$ denotes the temperature, $k > 0$ is the thermal conductivity, and f is a constant source term.

The boundary conditions are determined according to the procedure given in Code 2 of Appendix A.2. One of the following cases is selected:

$$\text{Case 1: } \left. \frac{dT}{dx} \right|_{x=0} = 0, \quad T(1) = 0,$$

$$\text{Case 2: } T(0) = 0, \quad T(1) = 0,$$

$$\text{Case 3: } \left. \frac{dT}{dx} \right|_{x=0} = 0, \quad \left. \frac{dT}{dx} \right|_{x=1} = 0.$$

Using the finite difference formulation specified in the assignment, with 11 grid points, the problem is represented by the linear system

$$MT = s, \quad (2)$$

where M is the matrix associated with the discretized differential operator, s is the source vector including the boundary-condition contributions, and T is the vector of unknown temperatures.

The objectives are:

1. Generate M and s using the student's enrollment number and solve the resulting linear system.
2. Present the numerical temperature distribution $T(x)$.
3. Plot the temperature distribution and interpret the result from a physical perspective.

A Generation Codes

The following codes reproduce the procedures provided in the assignment for generating the numerical data used in the computational experiments.

A.1 Code 1: Matrix and Vector Generation

Listing 1: Generation of the matrix A and vector b using the student's enrollment number.

```
import numpy as np

def gerar_matriz(matricula , n=10):
    np.random.seed(matricula)
    B = np.random.randint(-5, 6, size=(n, n))
    A = B @ B.T + n * np.eye(n)
    return A

def gerar_vetor(matricula , n=10):
    np.random.seed(matricula + 1)
    return np.random.randint(-20, 21, size=n)

matricula = 31837 # student's enrollment number

A = gerar_matriz(matricula)
b = gerar_vetor(matricula)

print("Matriz A:")
print(A)

print("\nVetor b:")
print(b)
```

A.2 Code 2: Heat Conduction Problem

Listing 2: Generation of the finite difference system for the one-dimensional heat conduction problem.

```
import numpy as np

def gerar_matriz_por_matricula(matricula , n=11):
    np.random.seed(matricula)

    h = 1 / (n - 1)
    M = np.zeros((n, n))
```

```

k = np.random.uniform(0.05, 0.3)
fonte = np.random.uniform(0.5, 2.0)

tipo_bc = np.random.choice([
    "neumann-dirichlet",
    "dirichlet-dirichlet",
    "neumann-neumann"
])

coef = 1 / h**2
s = np.ones(n) * (h**2 / k) * fonte

for i in range(1, n - 1):
    M[i, i - 1] = coef
    M[i, i] = -2 * coef
    M[i, i + 1] = coef

if tipo_bc == "neumann-dirichlet":
    M[0, 0] = -coef
    M[0, 1] = coef
    M[-1, -1] = 1
    s[-1] = 0

elif tipo_bc == "dirichlet-dirichlet":
    M[0, 0] = 1
    M[-1, -1] = 1
    s[0] = 0
    s[-1] = 0

elif tipo_bc == "neumann-neumann":
    M[0, 0] = -coef
    M[0, 1] = coef
    M[-1, -2] = -coef
    M[-1, -1] = coef

return M, s, k, fonte, tipo_bc

```

matricula = 31837 # student's enrollment number

```
M, s, k, fonte, bc = gerar_matriz_por_matricula(matricula)
```

```
print("Matriz_M:\n", M)
```

```
print("Vetor_s:\n", s)
```

```
print("k=", k)
```

```
print("fonte=", fonte)
```

```
print("BC=", bc)
```